Data Handling: Import, Cleaning and Visualisation

Lecture 11:
Wrap up

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Plan for today

1. Recap
2. Example of a Shiny app
3. Wrap up
4. Exam Info
5. Suggested Improvements
6. Concluding remarks
7. Q&A

Recap: Data Visualization and Text Data

Data visualization

Two ways: display data through *tables* or *graphs*.

Depends on the purpose.

`ggplot2` basics

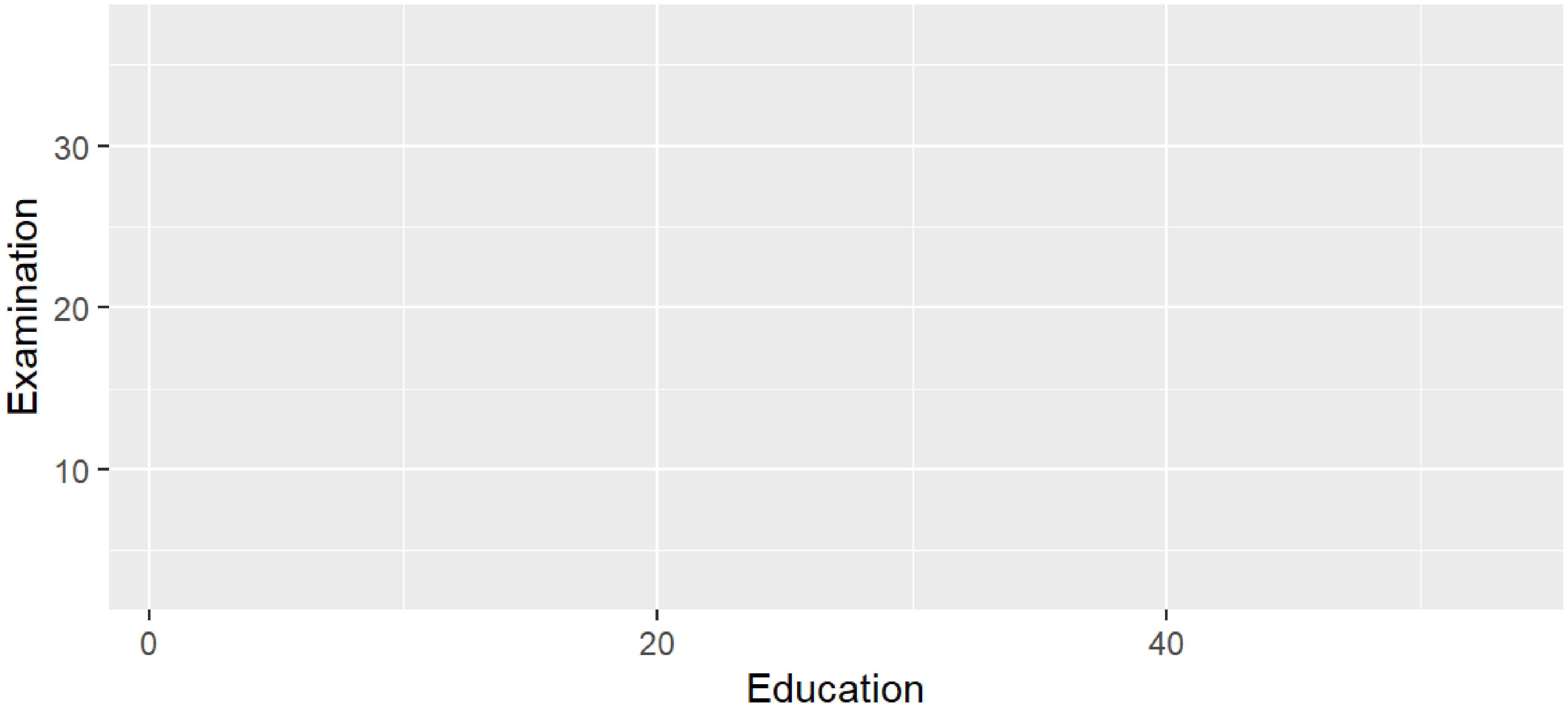
Using `ggplot2` to generate a basic plot in R is quite simple. Three key points:

1. The data must be stored in a `data.frame`/`tibble` (in tidy format!).
2. The starting point of a plot is always the function `ggplot()`.
3. The first line of plot code declares the data and the 'aesthetics' (e.g., which variables are mapped to the x-/y-axes):

```
ggplot(data = my_dataframe, aes(x= xvar, y= yvar))
```

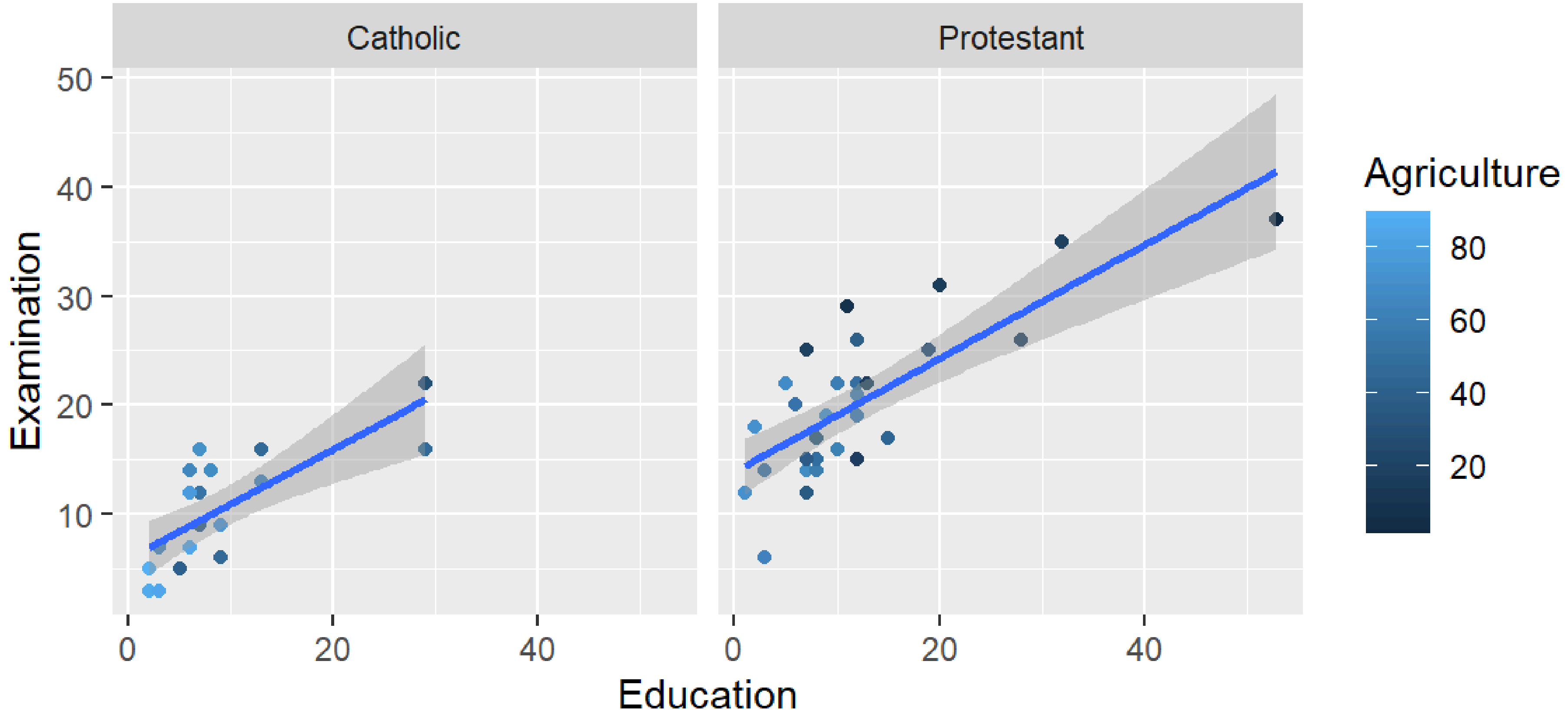
ggplot2: building plots layer by layer

```
ggplot(data = swiss, aes(x = Education, y = Examination))
```



ggplot2: building plots layer by layer

```
ggplot(data = swiss, aes(x = Education, y = Examination)) +
  geom_point(aes(color = Agriculture)) +
  geom_smooth(method = 'lm') +
  facet_wrap(~Religion)
```



Tufte's principles of data visualization

Minimize the Lie Factor

$$\frac{\text{size of effect shown in graphic}}{\text{size of effect in data}}$$

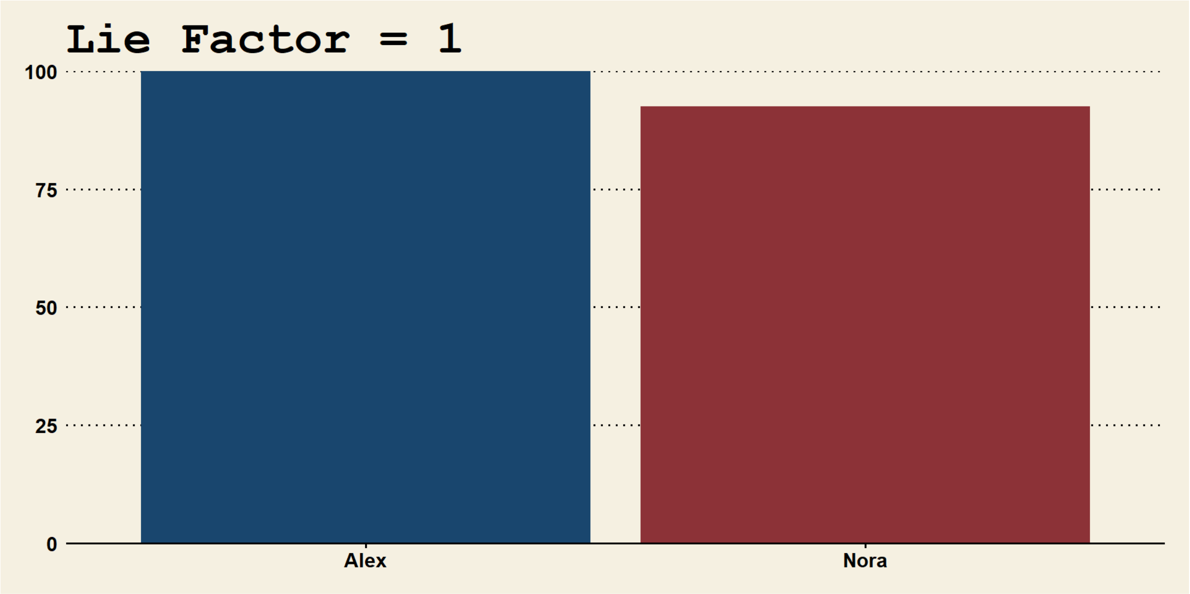
Maximize the Data-ink Ratio

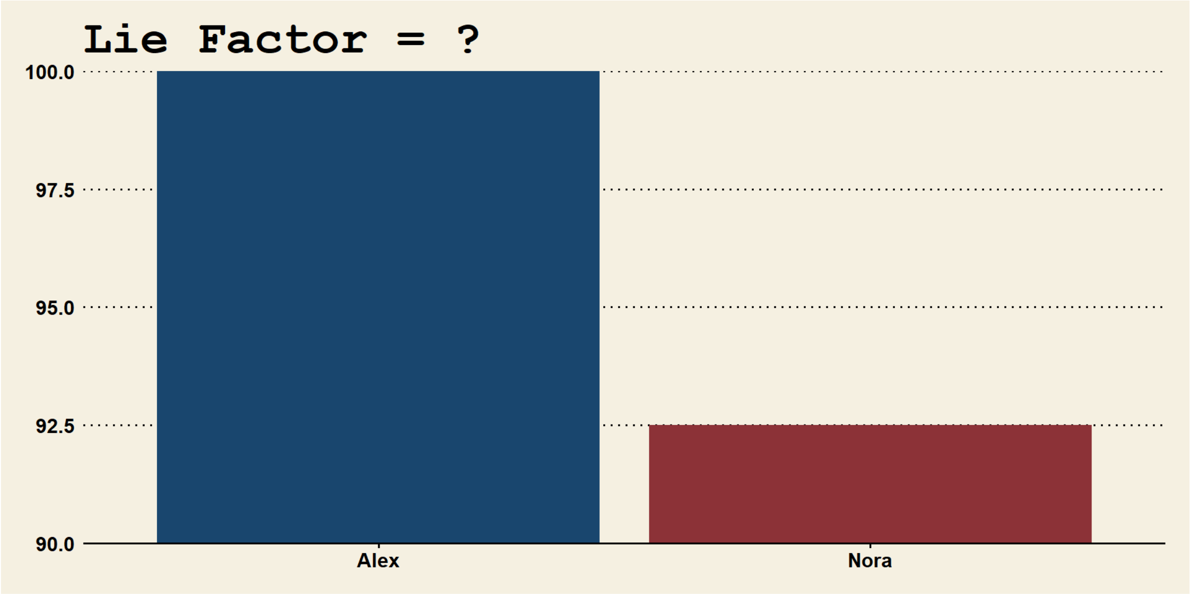
$$\frac{\text{ink used for data points}}{\text{total ink used to print the graphic}}$$

Maximize the Data density

$$\frac{\text{number of entries in data matrix}}{\text{area of data graphic}}$$

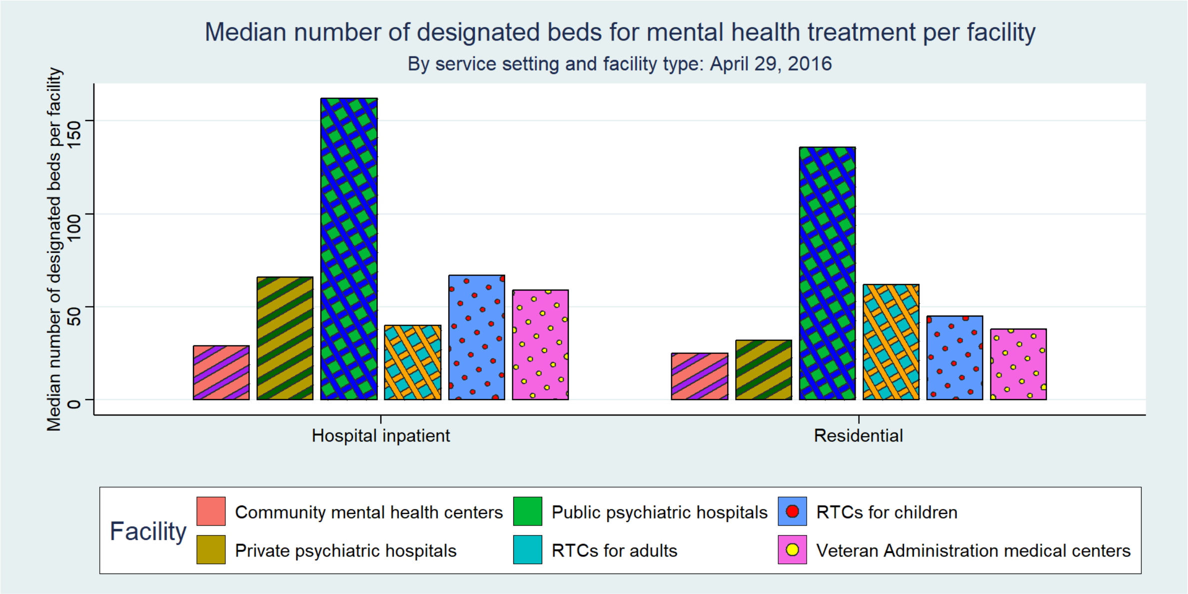
Minimize the Lie Factor

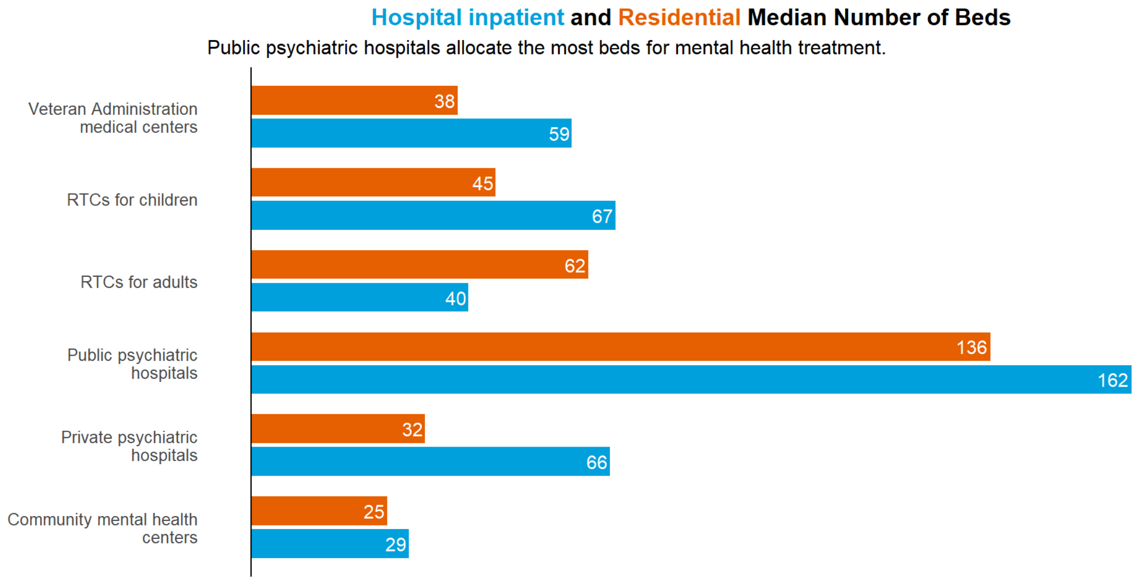




- Lie Factor

Maximize the Data-ink Ratio





▶ Show code for the graphs

▶ Show code for the graphs

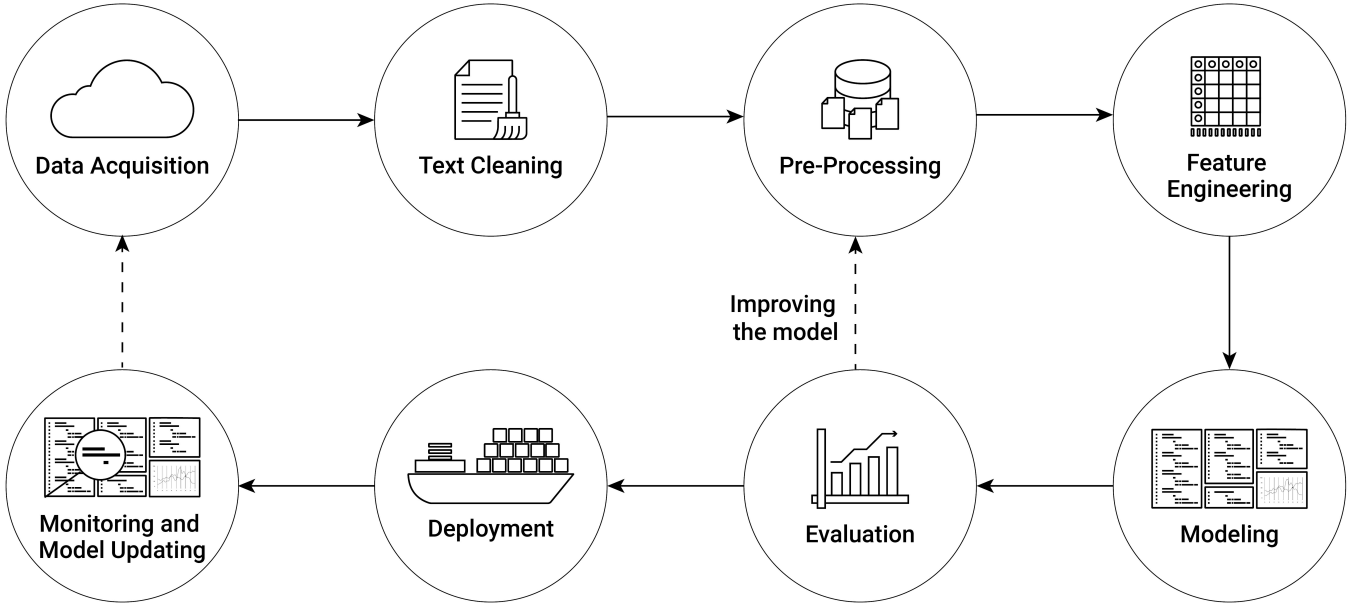
Maximize data density

Source: data-viz-workshop-2021

4 data points: 0.39, 0.56, 0.3, 2000 😱

Eight Steps in Text Analysis

Focus on steps 1-4 in this course.



- In R, we use `quanteda`: Comprehensive package for preprocessing, visualization, and statistical analysis of text data.

Data aqcuisition: From Raw Text to Corpus

The base, raw material, of quantitative text analysis is a **corpus**. A corpus is, in NLP, *a collection of authentic text organized into datasets*.

- In `quanteda`: A data frame with a character vector for documents and additional metadata columns.

```
print(corp)
```

```
Corpus consisting of 5 documents and 3 docvars.
text1 :
"Fellow-Citizens of the Senate and of the House of Representa..."

text2 :
"Fellow citizens, I am again called upon by the voice of my c..."

text3 :
"When it was first perceived, in early times, that no middle ..."

text4 :
"Friends and Fellow Citizens: Called upon to undertake the du..."
```

Cleaning/pre-processing: from corpus to tokens

Tokens: Building blocks of text analysis, smallest unit of text.

- The tokenization process allows to convert raw text into numerical representations by mapping each token to an integer ID.
  - e.g., words: “The cat sat” becomes a vector of three characters [“The”, “cat”, “sat”]
- Remove “noise” such as punctuation and stopwords.
- Create N-grams (e.g., bigrams like “not friendly”).

```
 [1] "Fellow-Citizens" "Senate"          "House"           "Representatives" "Among"
 [6] "vicissitudes"    "incident"        "life"            "event"           "filled"
[11] "greater"         "anxieties"       "notification"    "transmitted"     "order"
[16] "received"        "14th"            "day"             "present"         "month"
```

Feature engineering: From Tokens to Document-Term Matrix

- **DTM**: Document-Term Matrix with rows = documents and columns = tokens.
- Contains count frequencies or indicators.
- Sparse (most entries are zero). The dimensions need to be reduced.

```r
dfmat <- dfm(toks)
dfmat <- dfm_trim(dfmat, min_termfreq = 2) # remove tokens that appear less than 1 times
print(dfmat)
```

```
Document-feature matrix of: 5 documents, 602 features (56.18% sparse) and 3 docvars.
       features
docs    fellow-citizens senate house representatives among life event greater order day
  text1               1      1     2               2     1    1     2       1     2   2
  text2               0      0     0               0     0    0     0       0     0   0
  text3               3      1     0               2     4    2     0       0     4   1
  text4               2      0     0               0     1    1     0       1     1   1
  text5               0      0     0               0     7    2     0       0     3   1
[ reached max_nfeat ... 592 more features ]
```

Analyzing DTMs

Very basic statistics about documents are the **top features** of each document, the frequency of expressions in the corpus.

The frequency of tokens can be represented in a text plot.

```
library(quanteda.textplots)
quanteda.textplots::textplot_wordcloud(dfmat, max_words = 100)
```

Today

Shiny app

- **Shiny** is an R package that makes it easy to build interactive web apps straight from R.
- Example with Data from the European Quality of Government Index (EQI).
- This index focuses on both perceptions and experiences with public sector corruption.
- Data: **EQI-CATI Country Level 2010-2024** (only phone interviews).
- Source: QoG Institute

Data: EQI-CATI Country Level 2010-2024

- 28 Countries for 5 years (2010, 2013, 2017, 2021, 2024)
- Focus on three variables about the health care system:
  - `Ed_qual`/`Hel_qual`/`Law_qual`: *"How would you rate the quality of the education/public health care system/police force in your area?"* (1 very poor - 10 excllent)
  - `Helimpart1`: *"Certain people are given special advantages in the public health care system in my area."* (1 strongly disagree - 10 strongly agree)
  - `Hel_ask`: "*In the last 12 months, have you or anyone in your family been asked by a public official to give an informal gift or bribe in health or medical services?*" (Share of population who said "Yes" to above-stated question).

Data: EQI-CATI Country Level 2010-2024

```r
eqi <- read_csv("C:/Users/aurel/OneDrive/Documents/DataHandling/datahandling-lecture/materials/code_examples/qog_eqicati_long
  select(cname, year, Ed_qual, Hel_qual, Law_qual, Helimpart1, Hel_ask)

head(eqi)
```

```
# A tibble: 6 × 7
  cname    year Ed_qual Hel_qual Law_qual Helimpart1  Hel_ask
  <chr>   <dbl>   <dbl>    <dbl>    <dbl>      <dbl>    <dbl>
1 Austria  2010    6.24     6.81     6.64       4.27 NA
2 Austria  2013    6.51     6.80     6.53       5.07 NA
3 Austria  2017    6.68     7.27     7.22       5.72  0.0138
4 Austria  2021    7.19     7.67     7.42       4.85  0.00970
5 Austria  2024    6.90     6.92     7.16       5.47  0.0176
6 Belgium  2010    6.97     7.71     6.86       4.88 NA
```

Wrap up

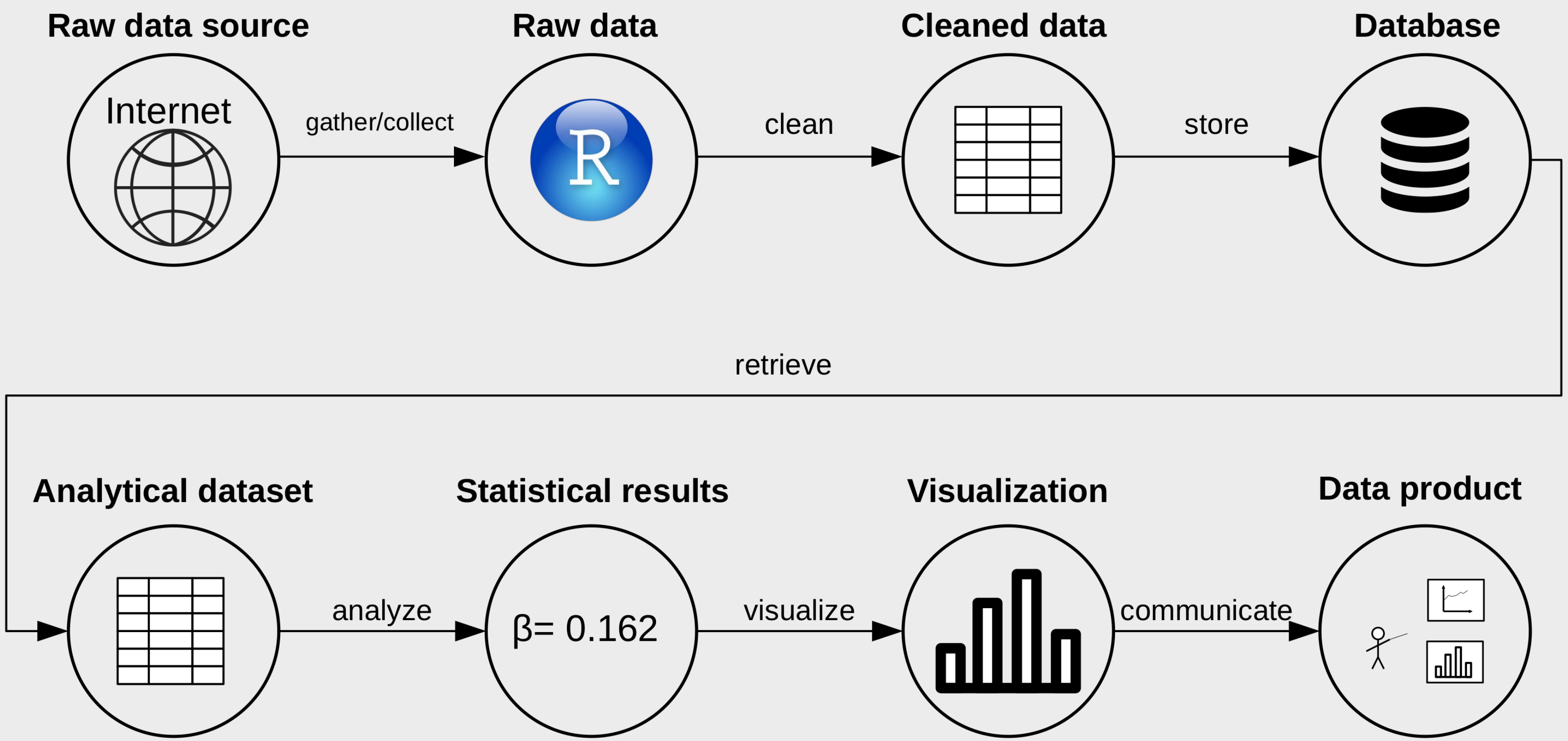
At the end of the course, you will be able to…

- Understand the tools you need when working with data
- Work independently with data
- Ask the right questions to a dataset
- Learn to communicate about data

My commitment to these goals and to your learning process

- Transferrable skills
- Hands-on approach
- Emphasis on real-world relevance
  (caveat: this course is mandatory for Econ students, I have limited freedom in the syllabus)
- As much fun as possible (as coding can be fun…😎)

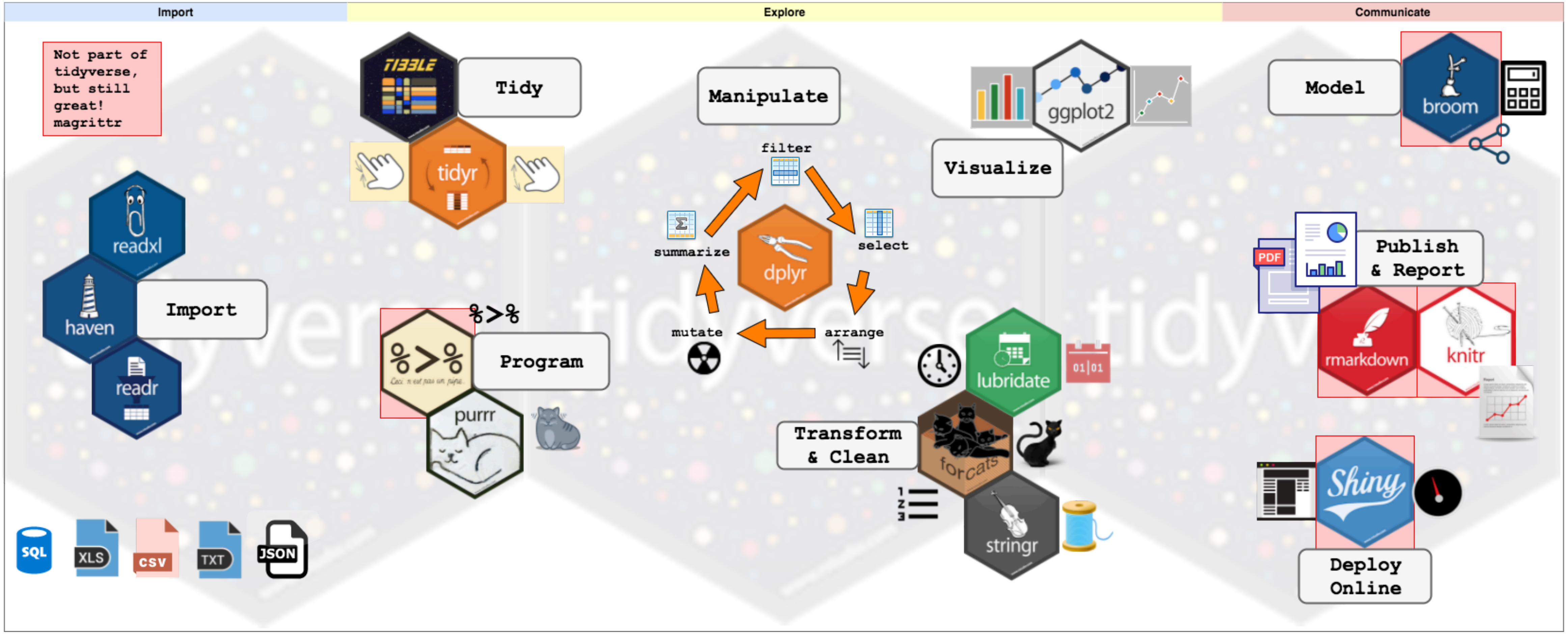
Data (science) pipeline



Wrap up: Conceptual part

- Understand the very basics of how computers process data.
  - Binary code.
  - Representation of binary code as text (encodings/standards!).
  - Text files to store data: special characters (comma, semicolon, etc.) define the structure (following a standard!)
    - CSV: two-dimensional/table-like structure
    - JSON/XML: hierarchical data (high-dimensional)
  - Data in text file: how is data structured when stored on the hard disk (mass storage device).
  - Data structures in R (objects): How data is structured/represented when loaded into RAM (via a ‘parser’)

Wrap up: Applied part: import, cleaning, analysis/visualisation.



Source: https://www.storybench.org/wp-content/uploads/2017/05/tidyverse.png

Wrap up: Applied part: import, cleaning, analysis/visualisation.

- Get from the data source to the final data product:
  - `tidyverse`: tools to help you with every part of the data pipeline.
- Import data into R. What to do if the parsing fails?
- Clean/prepare data in R. Aim: tidy data set (rows:observations, columns:variables).
- Filter, select, mutate, summarise.
- Compute basic summary statistics using group_by and summarise.
- Visualize raw data and basic statistics.

Exam

📢 Decentral exam and LockDown browser

Exam for exchange students

- 19.12.2024 at 16:15 in room 01-207.
- **BYO + cord 🔌 + 🖱️**

Lockdown browser

- Exam and LockDown Browser: check Sharepoint on StudentWeb and test on Canvas.
  Password: DataHandling2023

Code questions

- True/False questions, multiple-choice questions, and multiple-correct options. These are designed to test your understanding of the material.
- ❗ Exactly same style and structure as quizzes and mock exam ❗
- We do not invent wrong function parameters or misspell function names etc to mislead you!
- 'Passive' knowledge of key R syntax and most important functions of core lecture contents are important!

Essay/open questions

- There will also be 3-4 essay-style questions aimed at evaluating your ability to apply your knowledge to new situations.
  - E.g., you might be asked to explain particular steps of the data analysis process in a given situation. You can use code, R concepts, or you can explain in plain English. The more precise the better.
- **You will not be required to write exact R code**, but you should be able to interpret and understand the code provided in the exam.
- I expect you to be familiar with all R commands and concepts covered in the lectures, exercises, in-class code, and additional practice exercises.
- The readings are not mandatory for the exam. The focus will be on the material discussed in class and during the exercises.

Course Evaluation and Feedback

Course Evaluation

Very good overall evaluation. Thanks!!! 😊 😊

Positive feedback

- Quizzes, warm-up, exercises are helpful.
- In-class coding.
- Slides are nice.

Negative feedback

- Slides are too minimalistic.
- The pace was too fast.
- Objectives were not well-defined and clarified.

Suggested Improvements

- **Learning Progress**
  - "*some lectures are a little slow (e.g. binary number systems) and others are really fast. it would be nice if the pace evened out a bit.*"
  - For some, the course is too fast, for some too slow (dsf program). Suggestions?

- **Course Objectives**
  - Clearer objectives and learning outcomes at the beginning of the course, clearer expectations for the exam. Suggestions?

Suggested Improvements

- **More quizzes, more old exams, more mock exams** : 😵
- **Recordings** of the class: not an option for me.
- **html/pdf** issues in the slides. Sorry for this! 😞
- **Examination**: will be improved.

Q&A

Questions?

Final Remarks

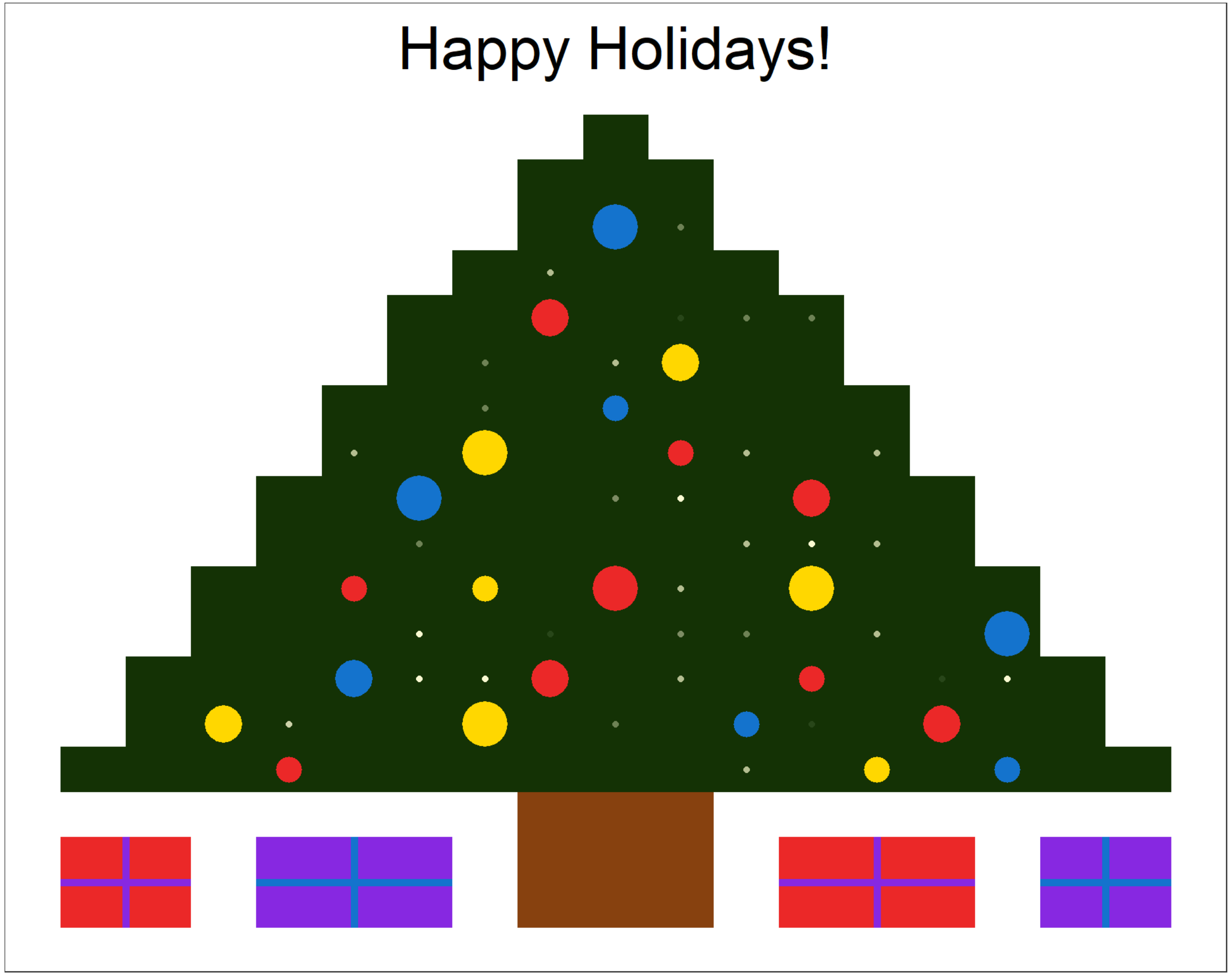
Final Remarks

Let's keep in touch!

- LinkedIn 📲
- Join the **R User Group in Zürich**!
  - Past events at Datahouse, SNB, Helsana, Swiss, etc.
  - Presentations from R-Users in the industry and in academia
  - Apéro + networking 🍕 🍺
  - Next event: **23.01.2025, 18:30 - 21:00** at the Swiss Federal Institute for Forest, Snow, and Landscape Research on Geodata and Landscape Insights with R.
  - Free Registration on Meetup

Final Remarks

- I had a lot of fun 🙏
- All the best for your exams! 👍
- All the best for your studies and careers, *and finally, of course, …*



Made with `ggplot`. Source: 'A very ggplot2 Christmas'